

Yinelemeli (Iterative) Algoritmalar

Analiz

Özyineli Algoritmalar(Recursive Algorithms)

- Analiz

Örnek:

$$T(n) = T(n - 1) + n$$

Çalışma zamanı analizini yapınız?

$$T(0) = 1$$

$$T(n) = T(n - 1) + n$$

$$= T(n - 2) + n - 1 + n$$

$$= T(n - 3) + n - 2 + n - 1 + n$$

...

$$= T(0) + 1 + 2 + 3 + \dots + n - 2 + n - 1 + n =$$

$$= 1 + 1 + 2 + 3 + \dots + n - 2 + n - 1 + n =$$

$$= 1 + \frac{n(n + 1)}{2}$$

$$T(n) = 2T(n-1) + 1$$

$$T(0) = 0$$

$$T(n) = 2T(n-1) + 1 = 2(2T(n-2) + 1) + 1$$

$$= 4T(n-2) + 2 + 1$$

$$= 8T(n-3) + 4 + 2 + 1$$

$$= 16T(n-4) + 8 + 4 + 2 + 1$$

$$= \dots$$

$$= 2^k T(n-k) + 2^{k-1} + 2^{k-2} + \dots + 8 + 4 + 2 + 1$$

HANOI KULELERİ

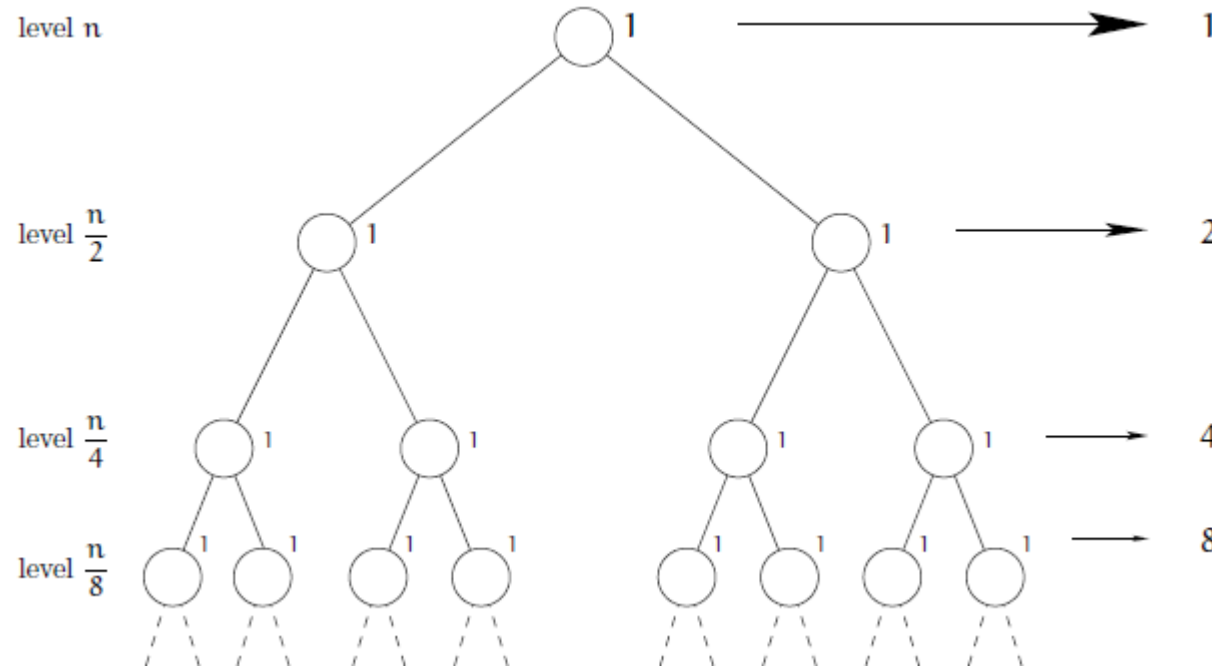
$$= 2^k T(n-k) + 2^{k-1} + 2^{k-2} + \dots + 8 + 4 + 2 + 1$$

$$= 2^k T(n-k) + \sum_{k=0}^{\infty} 2^{k-1}$$

$$= 2^k T(0) + 2^k - 1 = 2^n + 2^n - 1 = 2^{n+1} - 1$$

$$T(n) = 2T\left(\frac{n}{2}\right) + 1$$

$$T(1) = \Theta(1)$$



$$T(n) = \underbrace{1 + 2 + 4 + 8 + \dots}_{(\lg n + 1)}$$

$$= 1 + 2 + 4 + 8 + \dots + \frac{n}{2} + n$$

$$= \sum_{i=0}^{\lg n} \frac{n}{2^i} = n \left(\sum_{i=0}^{\lg n} \frac{1}{2^i} \right) \leq n \underbrace{\left(\sum_{i=0}^{\infty} \frac{1}{2^i} \right)}_2 = 2n$$

$$T(n) = \Theta(n)$$

Hızlı Sıralama

- Worst Case

$$T(n) = T(k) + T(n-k) + \alpha n$$

worst case

$$T(n) = T(1) + T(n-1) + \alpha \cdot n$$

$$T(n) = T(n-1) + T(1) + \alpha \cdot n$$

$$= T(1) + T(n-2) + \alpha(n-1) + T(1) + \alpha \cdot n$$

$$T(n) = T(n-2) + 2T(1) + \alpha(n-1 + n)$$

$T(n-2)$ yeniden yazılabilirse

$$T(n) = T(1) + T(n-3) + \alpha(n-2) + 2T(1) + \alpha(n-1 + n)$$

$$T(n) = T(n-3) + 3T(1) + \alpha(n-2 + n-1 + n)$$

$$T(n) = T(1) + \underline{T(n-4)} + \alpha \cdot (n-3) + 3T(1) + \alpha(n-2 + n-1 + n)$$

$$= T(n-4) + 4T(1) + \alpha[(n-3) + (n-2) + (n-1) + n]$$

Aynı biçimde devam edilerek i . adım için

$$T(n) = T(n-i) + iT(1) + \alpha[(n-i+1) + (n-i+2) + \dots + (n-1) + n]$$

$$T(n) = T(n-i) + iT(1) + \alpha \left[\sum_{j=0}^{i-1} n-j \right]$$

Burada dikkat edilirse i değeri en fazla $(n-1)$ 'e kadar gidebilir. Çünkü $(n-i)$ değeri 1'den büyük olamaz.

$T(n) = T(n-i) + i T(1) \dots$ O halde denklemi yeniden yazacak olursak ($i = n-1$ yazarak)

$$T(n) = T(n-(n-1)) + (n-1)T(1) + \alpha \sum_{j=0}^{n-1-1} n-j$$

$$T(n) = T(1) + (n-1)T(1) + \alpha \left[\sum_{j=0}^{n-2} (n-j) \right]$$

$$= nT(1) + \alpha \left[n^2 - 2n - \frac{1}{2}n^2 + \frac{3}{2}n - 1 \right]$$

$$T(n) = T(1) + (n-1)T(1) + \alpha \left[\sum_{j=0}^{n-2} n - \sum_{j=0}^{n-2} j \right]$$

$$= \overbrace{T(1) + (n-1)T(1)} + \alpha \left[n(n-2) - \frac{(n-2)(n-1)}{2} \right]$$

$$= \overbrace{nT(1)}^{\text{Değişken}} + \underbrace{\alpha}_{\text{Sabit}} \left[\overbrace{\frac{1}{2}n^2 - \frac{1}{2}n - 1}^{\text{Kare ve Sabit}} \right]$$

HızlıSıralama

- Best Case

$$T(n) = T(k) + T(n-k)$$

$$k = \frac{n}{2}, \quad n-k = n - \frac{n}{2} = \frac{n}{2}$$

"Best Case"

$$T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n}{2}\right) + \alpha n = 2 \cdot T\left(\frac{n}{2}\right) + \alpha \cdot n \quad T\left(\frac{n}{2}\right) = 2 \cdot T\left(\frac{n}{4}\right) + \alpha \cdot \frac{n}{2}$$

$$T(n) = 2 \cdot \left[2 \cdot T\left(\frac{n}{4}\right) + \alpha \cdot \frac{n}{2} \right] + \alpha \cdot n = 2^2 T\left(\frac{n}{4}\right) + 2 \alpha \cdot n$$

$$\left\{ \text{Burada } T\left(\frac{n}{4}\right) = 2 \cdot T\left(\frac{n}{8}\right) + \alpha \frac{n}{4} \right\}$$

yerine yazılırsa

$$T(n) = 2^2 \left[2 \cdot T\left(\frac{n}{8}\right) + \alpha \frac{n}{4} \right] + 2 \cdot \alpha n$$

$$= 2^3 T\left(\frac{n}{8}\right) + 3 \alpha \cdot n$$

"Best Case"

aynı biçimde devam edilirse, k . adımda

$$T(k) = 2^k T\left(\frac{n}{2^k}\right) + k \cdot \alpha n$$

$T(1)$ olması için rekursiyon ancak $n=2^k$ ya kadar gidi.

Diğer durumlarda $\frac{n}{2^k} < 1$ olur.

Bu durumda $n = 2^k$ olacaktır. $\log_2 n = k$

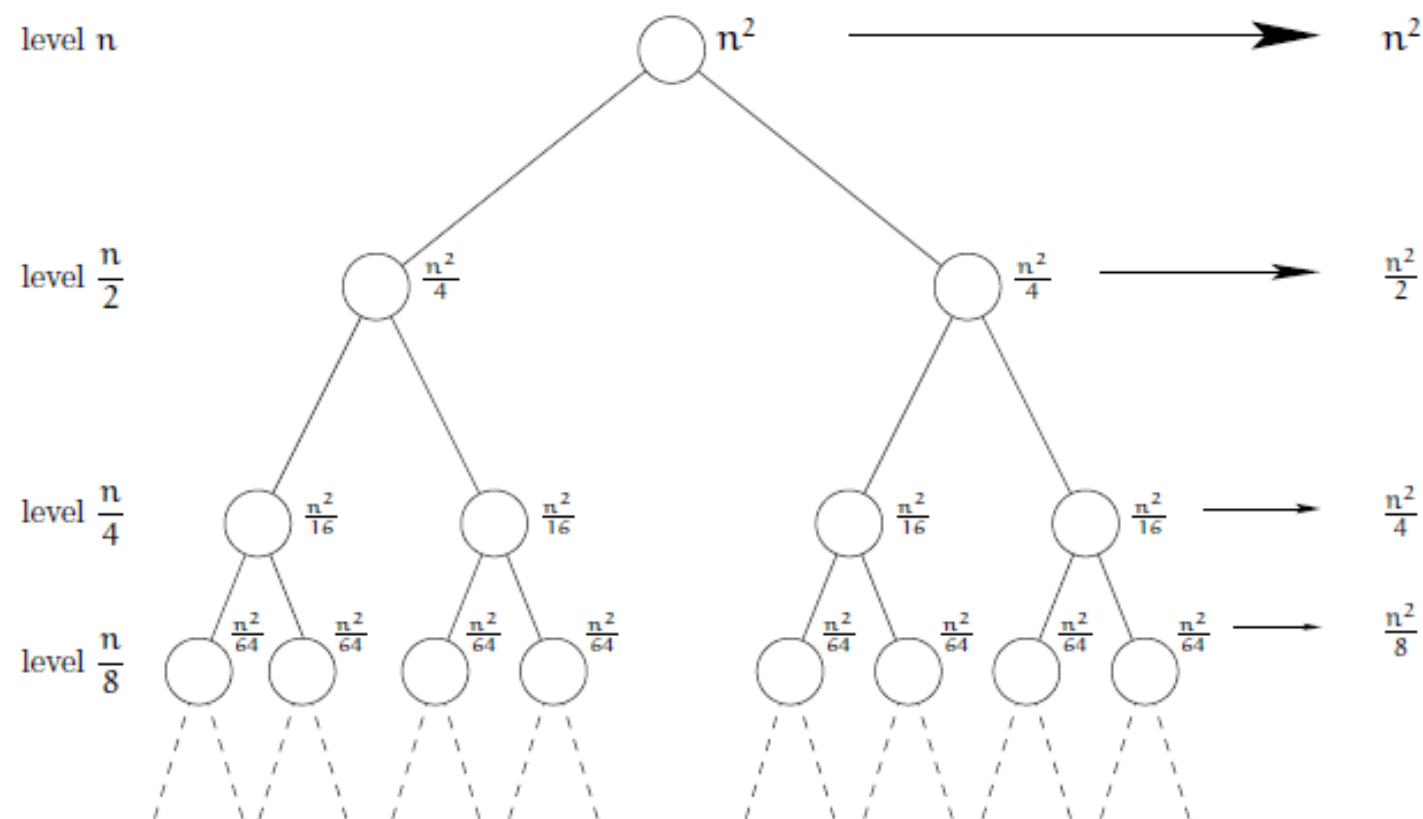
Denklemleri bu yeni duruma göre yeniden yazarsak

$$T(n) = 2^{\log_2 n} T\left(\frac{n}{2^{\log_2 n}}\right) + \log_2 n \cdot \alpha \cdot n$$

$$= n T(1) + \alpha \cdot n \cdot \log n \rightarrow O(n \cdot \log n)$$

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2$$

recursion tree



$$n^2 + \frac{n^2}{2} + \frac{n^2}{4} + \frac{n^2}{8} + \dots \leq n^2 \sum_{i=0}^{\infty} \frac{1}{2^i} = 2n^2$$

$$T(n) = \Theta(n^2)$$

$$T(n) = T(n-1) + \frac{1}{n}$$

$$T(n) = T(n-1) + \frac{1}{n}$$

$$= T(n-2) + \frac{1}{n-1} + \frac{1}{n}$$

$$= T(n-3) + \frac{1}{n-2} + \frac{1}{n-1} + \frac{1}{n}$$

...

$$= T(1) + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n-2} + \frac{1}{n-1} + \frac{1}{n}$$

$$= T(1) - 1 + \underbrace{1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n-2} + \frac{1}{n-1} + \frac{1}{n}}_{\Theta(\lg n)}$$

$$= O(1) + \Theta(\lg n)$$

$$= \Theta(\lg n)$$

$$T(n) = aT(n/b) + cn^d \quad (\text{for } n > 1)$$

$$a \geq 1, b \geq 2, c > 0 \text{ and } d \geq 0$$

$T(n)$ is $\Theta(n^d)$ if $a < b^d$

$T(n)$ is $\Theta(n^d \log n)$ if $a = b^d$

$T(n)$ is $\Theta(n^{\log_b a})$ if $a > b^d$.

$$T(n) = aT(n/b) + cn^d \quad (\text{for } n > 1)$$

$$\begin{aligned} T(n) &\text{ is } \Theta(n^d) \text{ if } a < b^d \\ T(n) &\text{ is } \Theta(n^d \log n) \text{ if } a = b^d \\ T(n) &\text{ is } \Theta(n^{\log_b a}) \text{ if } a > b^d. \end{aligned}$$

$$a = 2 \text{ and } b = 4, \text{ and } \log_4(2) = 1/2.$$

$$T(n) = 2T(n/4) + 1.$$

$$T(n) = \Theta(n^{1/2})$$

$$T(n) = 2T(n/4) + \sqrt{n}.$$

$$\sqrt{n} = \Theta(n^{1/2})$$

$$T(n) = \Theta(n^{1/2} \log_2(n)).$$

$$T(n) = 2T(n/4) + n$$

$$T(n) = \Theta(n)$$

$$T(n) = 2T(n/4) + n^2.$$

$$T(n) = \Theta(n^2)$$

$$T(n) = aT(n/b) + cn^d \quad (\text{for } n > 1)$$

$$T(n) \text{ is } \Theta(n^d) \text{ if } a < b^d$$

$$T(n) \text{ is } \Theta(n^d \log n) \text{ if } a = b^d$$

$$T(n) \text{ is } \Theta(n^{\log_b a}) \text{ if } a > b^d.$$

$$a. \ T(n) = 2T(n/2) + n^4.$$

$$b. \ T(n) = T(7n/10) + n.$$

$$T(n) = \Theta(n^4)$$

$$T(n) = \Theta(n).$$

$$T(n) = aT(n/b) + cn^d \quad (\text{for } n > 1)$$

$$T(n) \text{ is } \Theta(n^d) \text{ if } a < b^d$$

$$T(n) \text{ is } \Theta(n^d \log n) \text{ if } a = b^d$$

$$T(n) \text{ is } \Theta(n^{\log_b a}) \text{ if } a > b^d.$$

$$c. \ T(n) = 16T(n/4) + n^2.$$

$$d. \ T(n) = 7T(n/3) + n^2$$

$$T(n) = \Theta(n^2 \lg n)$$

$$T(n) = \Theta(n^2)$$

$$T(n) = aT(n/b) + cn^d \quad (\text{for } n > 1)$$

$$T(n) \text{ is } \Theta(n^d) \text{ if } a < b^d$$

$$T(n) \text{ is } \Theta(n^d \log n) \text{ if } a = b^d$$

$$T(n) \text{ is } \Theta(n^{\log_b a}) \text{ if } a > b^d.$$

$$T(n) = 7T(n/2) + n^2.$$

$$2T(n/4) + \sqrt{n}.$$

$$T(n) = \Theta(n^{\lg 7}).$$

$$T(n) = \Theta(\sqrt{n} \lg n)$$

Theorem 1 (Master Theorem, *Let $a \geq 1$ and $b > 1$ be constants, let $k = \lg_b a$, and let $f(n)$ be a positive function. Let*

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

$$T(1) = \Theta(1)$$

where $\frac{n}{b}$ is interpreted either as $\lfloor \frac{n}{b} \rfloor$ or $\lceil \frac{n}{b} \rceil$. Then $T(n)$ can be bounded asymptotically as follows.

Case 1 *If $f(n) = O(n^{k-\epsilon})$ for some positive constant ϵ then $T(n) = \Theta(n^k)$.*

Case 2 *If $f(n) = \Theta(n^k)$ then $T(n) = \Theta(n^k \lg n)$.*

Case 3 *If both*

- 1. $f(n) = \Omega(n^{k+\epsilon})$ for some positive constant ϵ , and*
- 2. $a.f\left(\frac{n}{b}\right) \leq c.f(n)$ for some constant c such that $0 < c < 1$ and for all sufficiently large n ,*

then $T(n) = \Theta(f(n))$.

Problem $T(n) = 3T\left(\frac{n}{4}\right) + n \lg n$

$$a = 3, b = 4, \quad \log_b a = \log_4 3 = 0.79$$

$$n \lg n = \Omega(n^{\log_4 3 + \epsilon})$$

$$0 < c < 1 \text{ and } 3\frac{n}{4} \lg \frac{n}{4} \leq cn \lg n \quad c = \frac{3}{4}, \quad \text{Case 3}$$

$$T(n) = \Theta(n \lg n)$$

1. $f(n) = \Omega(n^{k+\epsilon})$ for some positive constant ϵ , and
2. $a.f\left(\frac{n}{b}\right) \leq c.f(n)$ for some constant c such that $0 < c < 1$ and for all sufficiently large n ,

then $T(n) = \Theta(f(n))$.

Problem $T(n) = 2T\left(\frac{n}{2}\right) + n \lg n$

$$a = 2 \text{ and } b = 2, \quad \log_b a = \log_2 2 = 1 \quad n^1 = n$$

$$n \lg n = O(n^{1-\epsilon}) \quad \text{No}$$

$$n \lg n = \Theta(n^1)? \quad \text{No}$$

$$n \lg n = \Omega(n^{1+\epsilon}) \quad \epsilon > 0? \quad \text{No}$$

$$T(n) = \Theta(n \lg^2 n).$$

$$T(n) = 4T\left(\frac{n}{2}\right) + n$$

$$T(n) = 4T\left(\frac{n}{2}\right) + n^2$$

$$T(n) = 4T\left(\frac{n}{2}\right) + n^3$$

MasterTeoremi

Algoritma Analizi-Devam

$$T(n) = a T\left(\frac{n}{b}\right) + f(n) \quad a \geq 1 \quad b > 1$$

$$1. f(n) = O(n^{\log_b a - \epsilon}) \quad \epsilon > 0 \quad T(n) = \Theta(n^{\log_b a})$$

$$2. f(n) = \Theta(n^{\log_b a} \log^k n) \quad k \geq 0 \quad T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$$

$$3. f(n) = \Omega(n^{\log_b a + \epsilon}) \quad \epsilon > 0 \quad T(n) = \Theta(f(n))$$

$$af(n/b) \leq cf(n) \quad c < 1$$

$$T(n) = T\left(\frac{2n}{3}\right) + 1$$

$$T(n) = 1.T\left(\frac{n}{\frac{3}{2}}\right) + 1$$

$$a = 1$$

$$b = \frac{3}{2}$$

$$\log_b a = \log_{\frac{3}{2}} 1 = 0$$

Case 2

$$n^{\log_b a} = n^0 = 1 \quad f(n) = 1$$

$$1 = \Theta(n^0)$$

$$T(n) = \Theta(1 \cdot \lg n) = \Theta(\lg n)$$

$$T(n) = a T\left(\frac{n}{b}\right) + f(n)$$

$$T(n) = 16T(n/4) + n \implies (\text{Case 1}) \quad T(n) = \Theta(n^2)$$

$$T(n) = 2T(n/2) + n \log n \implies (\text{Case 2}) \quad T(n) = n \log^2 n$$

$$T(n) = T(n/2) + 2^n \implies (\text{Case 3}) \quad T(n) = \Theta(2^n)$$

$$T(n) = 64T(n/8) - n^2 \log n \quad \text{Master teoremi uygulanamaz} \quad (f(n) \text{ negatif})$$

$$T(n) = 6T(n/3) + n^2 \log n \implies (\text{Case 3}) \quad T(n) = \Theta(n^2 \log n)$$

$$T(n) = 3T(n/2) + n^2 \implies (\text{Case 3}) \quad T(n) = \Theta(n^2)$$

$$T(n) = 4T(n/2) + n^2 \implies (\text{Case 2}) \quad T(n) = \Theta(n^2 \log n)$$

$$T(n) = \sqrt{2}T(n/2) + \log n \implies (\text{Case 1}) \quad T(n) = \Theta(\sqrt{n})$$

$$T(n) = 0.5T(n/2) + 1/n \quad \text{Master teoremi uygulanamaz} \quad (a < 1)$$

$$T(n) = 2^n T(n/2) + n^n \quad \text{Master teoremi uygulanamaz} \quad a \text{ sabit değil}$$

MERGE SORT

$$T(n) = 2T(n/2) + \Theta(n)$$

altproblem sayısı *altproblem boyutu* *bölme ve birleştirme işi*

$$a = 2, b = 2$$

$$n^{\log_b a} = n^{\log_2 2} = n$$

$$(k = 0) \Rightarrow T(n) = \Theta(n \lg n) \quad (\text{Case 2})$$

$$(k = 0) \Rightarrow T(n) = \Theta(n \lg n)$$

$$T(n) = 1 T(n/2) + \Theta(1)$$

$$n^{\log_b a} = n^{\log_2 1} = n^0 = 1 \quad (k = 0) \quad (\text{Case 2})$$

$$T(n) = \Theta(\lg n)$$

